

Vanillin

Molecule behind the vanilla aroma

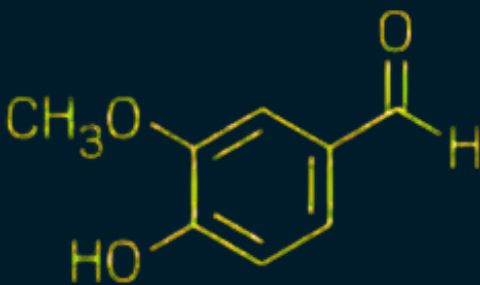
compound identity

Common name:
Vanillin

IUPAC name: 4-Hydroxy-3-methoxybenzaldehyde

Molecular formula:
 $C_8H_8O_3$

Molar mass: 152.15 g/mol



Important Properties

- White to slightly yellow crystalline solid
- Characteristic vanilla-like odor
- Contains one hydrogen-bond donor and three hydrogen-bond acceptors
- Moderately lipophilic compared with very small polar molecules
- Its functional groups contribute to its characteristic aroma and chemical behavior

Major Functional Groups

01

Phenol(Ar-OH)

02

Ether (-O-CH₃)

03

Aldehyde(-CHO)

04

Aromatic benzene ring

What Is Amazing About It?

A single relatively small molecule is strongly associated with the familiar vanilla aroma. Its aldehyde, phenolic hydroxyl, and methoxy groups work together with its aromatic structure to give vanillin its characteristic sensory properties.

Uses and Value

- Major flavoring compound associated with vanilla
- Used in foods and beverages
- Used in fragrances and flavor formulations
- Important commercial ingredient because it provides a recognizable vanilla flavor and aroma

Risk And Safety

- Avoid unnecessary exposure to concentrated vanillin.
- Handle laboratory-grade material according to its safety data sheet.
- Prevent contact with eyes and prolonged skin exposure.
- Follow appropriate laboratory handling and disposal procedures.

Chemical Engineering Connection

Vanillin can be connected to reaction engineering, separation, crystallization, and product formulation. Industrial production and purification require controlled processing and separation of the desired compound from other materials before it can be incorporated into food or fragrance products.

References

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